Statistics/Data analysis

## Title

**rdbw2d** —— Bandwidth selection for location-based boundary discontinuity designs.

## Syntax

**rdbw2d** *y x1 x2 assignment* [*if*] [*in*], **b(***# # [# # ...]***)** [**fuzzy(***varname***)** **p(***#***)**
**kernel(***kernel***)** **kerneltype(***type***)** **bwselect(***selector***)** **bwparam(***target***)**
**method(***method***)** **vce(***vcetype***)** **bwcheck(***#***)** **masspoints(***masspointsoption***)**
**cluster(***clustvar***)** **scaleregul(***#***)** **scalebiascrct(***#***)** **stdvars(***on*|*off***)**]

## Description

**rdbw2d** computes bandwidths for rd2d. The returned table has one row per boundary point and columns **b1**, **b2**, **h01**, **h02**, **h11**, **h12**, **N_Co**, and **N_Tr**.

## Options

**b()** specifies boundary evaluation points as pairs and is required.

**fuzzy()** specifies treatment receipt/status for fuzzy bandwidth selection. If supplied, **bwparam()** controls whether the selector targets the fuzzy Wald ratio or the reduced-form outcome discontinuity.

**p()** sets the local polynomial order. Default is **p(1)**.

**kernel()** specifies the kernel function. Available choices are:
- **triangular** or **tri**: triangular kernel, the default.
- **epanechnikov** or **epa**: Epanechnikov kernel.
- **uniform** or **uni**: uniform kernel.
- **gaussian** or **gau**: Gaussian kernel.

**kerneltype()** specifies the kernel structure. Available choices are:
- **prod**: product kernel, the default.
- **rad**: radial kernel.

**bwselect()** specifies the bandwidth selector. Available choices are:
- **mserd**: one common MSE-optimal bandwidth selector for the boundary RD treatment-effect estimator at each evaluation point, the default.
- **cerrd**: CER-optimal counterpart of **mserd**.
- **imserd**: IMSE-optimal common bandwidth selector for the boundary RD treatment-effect estimator based on all evaluation points.
- **icerrd**: CER-optimal counterpart of **imserd**.
- **msetwo**: two different MSE-optimal bandwidth selectors, one for the control group and one for the treated group, at each evaluation point.
- **certwo**: CER-optimal counterpart of **msetwo**.
- **imsetwo**: two IMSE-optimal bandwidth selectors, one for each group, based on all evaluation points.
- **icertwo**: CER-optimal counterpart of **imsetwo**.

**bwparam()** specifies the target parameter for fuzzy bandwidth selection. Available choices are:
- **main**: selects bandwidths for the linearized fuzzy Wald ratio, the default.
- **itt**: selects bandwidths for the reduced-form outcome discontinuity. This option is ignored in sharp designs.

**method()** specifies the bandwidth calculation method. Available choices are:
- **dpi**: data-driven plug-in MSE-optimal selector, the default.
- **rot**: rule-of-thumb selector.

**vce()** specifies the variance-covariance estimator used in the bandwidth calculations. Available choices are **hc0**, **hc1**, **hc2**, and **hc3**. Default is **hc1**.

**bwcheck()** enlarges preliminary bandwidths, when needed, so that at least the requested number of unique observations are included in each side-specific window. Default is **bwcheck(20)**.

**masspoints()** specifies how repeated running-variable values are handled. Available choices are:
**check**: detects mass points and reports unique-observation counts, the default.
**adjust**: adjusts preliminary bandwidths to ensure a minimum number of unique observations within each side of the boundary.
**off**: ignores mass points.

**cluster()** specifies a cluster ID variable for cluster-robust bandwidth calculations.

**scaleregul()** specifies the scaling factor for the regularization term in bandwidth selection. Default is **scaleregul(1).**

**scalebiascrct()** specifies the scaling factor used for bias correction based on higher-order expansions. Default is **scalebiascrct(1).**

**stdvars()** controls whether the running variables are standardized before computing automatic bandwidths. Use **stdvars(on)** or **stdvars(off).** Default is **on.**

**Examples**

```
. clear
. set obs 800
. set seed 123
. gen x1 = rnormal()
. gen x2 = rnormal()
. gen d = x1 >= 0
. gen y = 3 + 2*x1 + 1.5*x2 + d + rnormal()
. rdbw2d y x1 x2 d, b(0 0 0 1)
```

**Stored results**

**rdbw2d** stores **e(bws), e(N), e(p),** and option labels in **e().**